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ORIGINAL PAPER

Does Social Media Pressure Induce Corporate Hypocrisy? Evidence of ESG Greenwashing from China

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Abstract

Hypocrisy in corporate social responsibility has attracted increasing attention from scholars. Under the context of the digital era, we examine whether and how social media pressure affects the typical corporate hypocrisy of ESG greenwashing. Using a dataset of A-share listed firms in China from 2011 to 2021, we find that negative sentiment from stock message boards induces firms to engage in ESG greenwashing. Our results are confirmed by an array of robustness tests and additional tests. In addition, the impact of social media pressure is more pronounced in firms with lower market value and poorer reputation. The channel tests show that social media pressure induces ESG greenwashing by exacerbating managerial myopia. Finally, we further find that the shareholding of institutional investors, high-quality audit and local environmental regulation can alleviate this negative impact. Overall, our study demonstrates the mechanism underlying social media pressure on ESG greenwashing and provides practical suggestions for improving the construction of the ESG system.

Keywords Social media pressure · Corporate hypocrisy · ESG greenwashing

Introduction

Corporate social responsibility (CSR) has always faced criticism and scepticism. The proliferation of greenwashing has resulted in an increase in consumer cynicism and mistrust (Jahdi & Acikdilli, 2009). For example, in September 2022, Swedish clothing giant H&M, which operates globally on the basis of sustainability, fashion and profit, was accused by the Dutch Authority for Consumers and Markets (ACM) of making 'environmentally misleading claims'. In another example, Tesla (TSLA: NASDAQ), a global leader in new energy vehicles, was fined by the US Environmental Protection Agency (EPA) in February 2022 for violating exhaust gas control regulations and was removed from the S&P 500 ESG Index in May for not disclosing its low-carbon strategy. There is a large number of similar cases. Given this, it is widely believed that CSR activities are driven by self-interest (Mohr et al., 2001), and firms use CSR activities as a marketing strategy to enhance their corporate image, with the ultimate aim of increasing profits (Alhouti et al., 2016). Meanwhile, the firm's CSR claims are often inconsistent with their actual CSR behaviours (Cho et al., 2015; Jahdi & Acikdilli, 2009), and corporate hypocrisy such as greenwashing has become a phenomenon that cannot be ignored in CSR practice.[1] However, the literature on the determinants of corporate hypocrisy from the perspective of pressure generated by small investors, particularly via social media, remains scant. In fact, the rise of social media, especially Internet stock message boards, offers small investors new avenues via which to gather firm-relevant information and speak up for themselves. At this point, firms have, to a certain extent, lost control of their information environment (Miller & Skinner, 2015). Small investors actively speak through social media, which may generate sufficient pressure from public opinion to affect corporate management (Ang et al., 2021; Li & Zhang, 2023; Liu, 2022; Sun et al., 2022). However, as regards whether and how social media pressure

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[1] Prior literature defines corporate hypocrisy as 'the belief that a firm claims to be something that it is not' (Wagner et al.,2009), It is believed that 'inconsistency between word and deed' is the most obvious manifestation of corporate hypocrisy (Fassin and Buelens, 2011).

influences corporate hypocrisy, the existing literature does not provide consistent answers.

As a typical example of corporate hypocrisy in CSR practice, ESG greenwashing refers to the way that a firm selectively discloses large quantities of ESG data to obscure its less impressive overall ESG performance, thus proving that 'claims and actions do not meet' in ESG practice (Walker & Wan, 2012; Yu et al., 2020). We develop two competing hypotheses to explore whether and how ESG greenwashing is influenced by social media pressure. On the one hand, based on impression management theory (Goffman, 1959), public opinion pressure formed by investors' comments on social media can increase managers' motivation for engaging in impression management, thus inducing the myopic behaviour of ESG greenwashing (the 'impression management hypothesis'). On the other hand, based on principal–agent theory (Jensen & Meckling, 1976), small investors' active comments on social media can improve corporate governance and suppress managers' motivation for information manipulation, thus inhibiting ESG greenwashing (the 'governance improvement hypothesis').

To clarify the applicability of the above two hypotheses, we use a sample of Chinese listed firms over the period 2011–2021 and examine the impact of social media pressure on ESG greenwashing. China is an ideal market to test our question for the following reasons. First, in recent years, with the increasing attention of the Chinese government on environmental issues, ESG practices are booming in China,[2] and the ESG performance of firms has gradually become an important basis for investor decision-making (Cao et al., 2022). At the same time, ethical awareness regarding CSR in China is far from mature (Du, 2015). Chinese listed firms have a strong motivation to engage in ESG greenwashing. They may obtain high returns from the capital market through this low-cost impression management approach. Second, as a kind of voluntary information disclosure, ESG reports currently lack a well-developed information disclosure system, unified information disclosure norms and mature information disclosure authentication in China. Meanwhile, investors in China's capital market are mostly small investors (Ang et al., 2021), who are at an information disadvantage and lack the capacity for information analysis and processing (Malmendier & Shanthikumar, 2007), so it is difficult for them to detect corporate ESG greenwashing in an accurate and timely manner. Therefore, Chinese listed firms have ample opportunity and space for manipulative information to achieve ESG greenwashing. In our specific research design, we focus on the pressure caused by negative sentiment of small investors on stock message boards. The reasons are as follows. First, as an investment awareness, ESG is more concerned by investors than the public. In particular, China does not have social media platforms similar to Twitter and Seeking Alpha for investors sharing stock investment information (Ang et al., 2021). Therefore, a stock message board is a suitable and unique social media platform for testing our hypotheses. Second, compared to number of postings, sentiment analysis of postings more accurately describes investors' satisfaction with listed firms. Furthermore, compared to positive sentiment, negative sentiment in social media postings is more likely to be widely spread (Li et al., 2020),[3] and can generate a greater impact on corporate decision-making (Ang et al., 2021). To ensure the reliability of the data, following prior literature (Ang et al., 2021; Huang et al., 2016; Li & Zhang, 2023), we construct an indicator of social media pressure (negative sentiment) based on postings of *EastMoney*, the most influential financial social media platform in China. We find that social media pressure can induce firms to conduct ESG greenwashing by exacerbating managerial myopia, which supports the impression management hypothesis. Moreover, this effect is more pronounced in firms with higher market value pressure and poorer reputation, while institutional investors participation, high-quality audit and local environmental regulation can mitigate this negative impact.

We make contributions to three strands of literature. First, the paper adds to the literature on the determinants of corporate hypocrisy, especially that focusing on ESG greenwashing. Prior literature on corporate hypocrisy examines the attribution of perceptions of corporate hypocrisy (Demasi & Voegtlin, 2023; Janney & Gove, 2011; Lauriano et al., 2022) and attempts to find corresponding avoidance strategies (Bellucci et al., 2021; Carlos & Lewis, 2018; Wagner et al., 2009). In addition, some literature explores the consequences of corporate hypocrisy (Du et al., 2023; Marín et al., 2016). With increasing ESG awareness, a growing body of literature investigates the determinants of environmental hypocrisy (i.e. greenwashing). Studies have so far examined the influence of internal characteristics such as corporate output growth and profits (Kim & Lyon, 2015), as well as external pressure from social media (Lyon & Montgomery, 2013), regulation (Kim & Lyon, 2015), stakeholders such as suppliers, customers, shareholders and industry associations (Testa et al., 2018), policy subsidies (Zhang,

[2] According to statistics from the Global Alliance for Sustainable Investment (GSIA), as of September 2023, the market size for major types of responsible investments in China was approximately CNY 31.59 trillion.

[3] Negativity bias theory in psychology shows that negative information has a processing advantage over positive information, people are more sensitive to negative stimuli, and the transmission of sentiment is asymmetric, and negative sentiment is more likely to lead to empathy and transmission (Baumeister et al., 2001; Bebbington et al.,2017).

2023a) and institutional environment (Marquis et al., 2016; Zhang, 2022, 2023c). However, there is not much empirical literature analysing how public opinion pressure influences greenwashing. In literature similar to the theme of this paper, Lyon and Montgomery (2013) theorise that the popularity of Facebook and Twitter can alleviate information asymmetry between consumers and companies, thereby reducing the incidence of greenwashing. But they also point out that empirical work testing this proposition would be highly worthwhile. Cao et al. (2022) suggest that Chinese firms with greater advertising (sales) expenses may engage in greenwashing by opportunistically utilising media, which is seen as a key channel for ESG image management. In contrast to the above literature, based on impression management theory, our empirical evidence suggests that negative sentiment from stock message boards induces ESG greenwashing by exacerbating managerial myopia. We further discuss the motivation for and governance of corporate ESG greenwashing.

Second, our study contributes to the literature on social media from the perspective of business ethics. Prior literature on social media mainly focuses on its functions of information exchange (Cade, 2018; Lee et al., 2015; Miller & Skinner, 2015) and information aggregation and prediction (Antweiler & Frank, 2004; Bartov et al., 2018; Tang, 2018). Some scholars acknowledge the importance of social media in improving the market information environment and corporate governance (Ang et al., 2021; Li & Zhang, 2023; Liu, 2022; Lyon & Montgomery, 2013; Sun et al., 2022). However, from the perspective of business ethics, we find the social media pressure may dissimilate the behaviour of managers, and induce myopia and opportunism. This conclusion reminds us that, while using social media for communication and governance, attention should also be paid to its negative impact on corporate management.

Third, we contribute to the literature on the impacts of small investors on corporate management. There is a long-standing debate regarding whether small investors positively or negatively affect corporate management. Some scholars argue that supervision from small investors can constrain value-destroying activities by managers or major shareholders and improve corporate governance (Ang et al., 2021; Chen et al., 2013; Fried et al., 2020; Li, 2021; Li & Zhang, 2023). Others contend that, owing to limitations such as information disadvantages, irrationality and short-term preferences, small investors may be unable to influence managerial decisions, or may even induce managerial myopia or a chaotic management style (Belloc, 2013; Cremers et al., 2020; Malmendier & Shanthikumar, 2007). We find that the negative sentiment from small investors may lead a firm to create a false impression to mislead investors about actual ESG performance, which supports the second viewpoint.

The remainder of this paper is organised as follows. "Theoretical Analysis and Hypothesis Development" section develops the theoretical framework and hypotheses. "Research Design" section elaborates research design, including the sample selection, variable construction and model. "Empirical Results" section presents descriptive statistics and the baseline results. "Robustness Tests" and "Additional Tests" provide robust evidence for the results. "Channel Tests" section demonstrates the influencing mechanism. "Cross-sectional Tests" and "Further Analyses" present several cross-sectional tests. "Conclusion and Discussion" section concludes.

Theoretical Analysis and Hypothesis Development

In recent years, the rise of social media has provided important assistance for small investors to exchange information with each other, form interest alliances and enhance market influence. Compared with traditional news media (e.g. television, print media and news websites), social media has advantages in shaping public opinion. First, the release of information on social media is subject to less control and scrutiny. This greatly enhances the inclusiveness of information content, which is more conducive to ordinary people speaking freely and forming a broader public opinion foundation. Traditional media, in contrast, is subject to strict factual verification and regulation (Lyon & Montgomery, 2013), and more importantly, is produced 'by and for elites' (Deephouse & Suchman, 2008). Second, social media has a higher communication efficiency because of strong interactivity. Compared with the 'one-to-many' information dissemination mode of traditional news media, the information dissemination mode of social media is 'many-to-many'; that is, multiple users publish information, and multiple users receive information. This renders it relatively easy for the public's attention on a specific event to experience 'exponential progression' growth, which can quickly form 'hotspots' and cause a great impact on public opinion. Furthermore, it is worth mentioning that, in recent years, the audience of traditional media represented by television and print media has been in significant decline.[4] Therefore, their influence in shaping and disseminating public opinion is greatly weakened. Third, social media has a higher information density. The playback duration or layout of traditional media is limited. In contrast, the scalability of the online information

[4] According to data from CSM Media Research (https://www.csm.com.cn/en/), the per capita viewing time of television viewers decreased by about 33% from 2013 to 2022. During the same period, statistics from the China Newspaper Association (CNA) showed that the total print volume of newspapers decreased by about 61%.

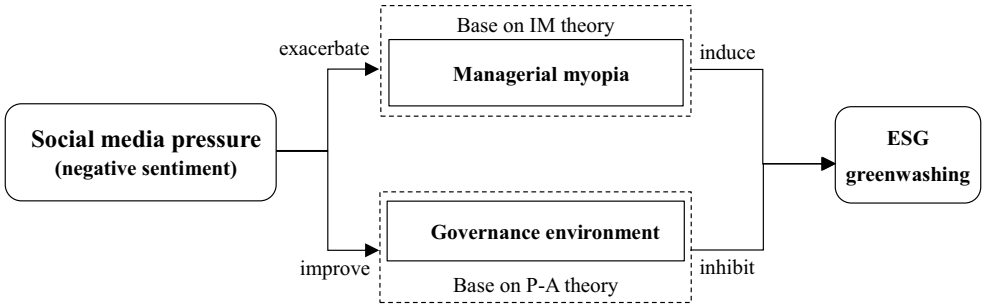


Fig. 1 Theoretical analysis framework

space enables social media to accommodate massive amounts of information, while lower levels of review and supervision greatly improve the speed of information generation and updating. Therefore, both information volume and update speed are significantly higher than for traditional news media. Finally, the participants of social media are more targeted. Taking stock message boards as an example, there is a separate forum for each listed firm on these boards, and most of the discussants in the forum are investors who hold or are preparing to hold the firm's stocks. Therefore, compared to traditional media users, they pay more attention to the firm and have more targeted information exchanges in forums.

The above characteristics of social media help to enhance the power of small investors and form strong public opinion pressure, thus affecting corporate management (Ang et al., 2021; Li & Zhang, 2023; Liu, 2022; Sun et al., 2022). Drawing on impression management theory and principal–agent theory, we analyse the effect of social media pressure on the typical hypocritical behaviours of firms (i.e. ESG greenwashing) and propose two competitive hypotheses, as outlined above. The theoretical framework of this paper is shown in Fig. 1.

Analysis Based on Impression Management (IM) Theory

Impression management refers to the behavioural strategies that people use to create desired social images or identities (Tetlock & Manstead, 1985). The literature on organisational impression management points out that firms can influence the response of external stakeholders to events by disclosing or not disclosing specific information (Elsbach et al., 1998). In anticipating that stakeholders may become dissatisfied with the firm due to some event, the firm releases strategic noise—news that is neither intended to clarify the initial event nor causally related to the initial event (Graffin et al., 2011)—in order to distract stakeholders from the focus event (Hirshleifer et al., 2009) and help offset the potential negative impact of this event (Graffin et al., 2016).

In recent years, with the growing popularity of the concept of sustainable development and the increasing attention of investors to corporate ESG, a high level of ESG information disclosure can have a beneficial effect on firm reputation and may act as ideal 'strategic noise' for firms when facing social media pressure. Prior studies find that outstanding ESG performance can decrease systematic risk (Albuquerque et al., 2019), send a positive signal of good performance to the outside world and win the trust of stakeholders (Christensen et al., 2022). A high level of ESG information disclosure can also have a positive impact on corporate competitiveness and corporate value (Qureshi et al., 2020; Raimo et al., 2021).

However, when negative sentiment prevails in social media, it is difficult for firms to increase their actual investment to improve ESG performance. Small investors who are active on social media pay more attention to firms' immediate performance owing to their low shareholding ratio and strong liquidity but have less understanding of corporate long-term strategies. It is well established that managers, especially those being pressured by the market, have a tendency to forgo the long-term interest to achieve short-term profits and gradually deviate from the goal of maximising the value of the firm (Li & Zhang, 2023). For example, one prior study finds that short-term performance pressure by investors can lead to a drop in the future innovation, a high-risk, slow-acting investment activity that is nonetheless conducive to the long-term development of a firm (Cremers et al., 2020). Similar to innovation, the ESG practices of firms also require investment in their early stages, and are thus conducive to long-term development but not short-term financial performance (Li et al., 2017a, 2017b). Therefore, to avoid exacerbating negative sentiment from social media due to short-term performance decline, firms may reduce ESG practices.

Compared with costly ESG practices, ESG greenwashing is a low-cost behavioural strategy with controllable risks and may therefore be a 'rational' choice for firms to conduct

impression management in the face of social media pressure. First, as an information manipulation behaviour, ESG greenwashing has almost no direct cost and does not affect short-term profits, which can effectively alleviate the short-term performance pressure arising from ESG practices. Second, as a voluntary initiative, ESG information disclosure does not have a widely implemented reporting framework, and firms have discretion in their choice of disclosure (Du & Yu, 2021). Walker and Wan (2012) also note that, in line with a lack of binding regulations on ESG information disclosure, management may adopt ESG information manipulation to hide the real investment cost of ESG. Third, ESG information has high emotional resilience and ambiguity, with obvious qualitative characteristics, which makes it difficult for small investors to accurately identify ESG greenwashing. Small investors on social media are at an information disadvantage, often lacking the ability and resources to analyse the intrinsic motivation of executive decision-making (Malmendier & Shanthikumar, 2007). Meanwhile, most people lack a direct understanding of the actual environmental quality of a company's production process, and it is challenging to accurately evaluate a company's environmental footprint (Lyon & Maxwell, 2011). Therefore, the risk of ESG information manipulation being exposed is greatly reduced. In fact, a series of recent papers also show that ESG greenwashing has indeed become a strategic behaviour of firms for coping with political, economic and environmental pressures (Yu et al., 2020; Zhang, 2022, 2023c) or achieving managers' self-interest (Cao et al., 2022). Drawing on impression management theory, Tata and Prasad (2015) suggest that the authenticity, integrity and reliability of social responsibility information is affected by management's impression management behaviour, which can mislead stakeholders in their decision-making. In the legitimacy-based research, firms are characterised as strategically using sustainability disclosures and reports for window-dressing and impression management purposes (Cho et al., 2015).

In addition, small investors' large-scale information aggregation, dissemination and exchange on stock message boards places the firm's behaviour in view of the public, which further affects the fluctuation of the firm's stock price (Antweiler & Frank, 2004). Furthermore, relative to positive information, negative information is more likely to be widely disseminated (Bebbington et al., 2017). Negative sentiment from social media is more likely to spread among investors (Li et al., 2020), thus posing a greater risk of stock price decline. Consequently, facing criticisms from social media, firms may also have stronger incentives to conduct ESG greenwashing.

Based on the above analysis, we propose Hypothesis H1a:

H1a Social media pressure induces firms to engage in ESG greenwashing.

Analysis Based on Principal–agent (P–A) Theory

According to principal–agent theory, because of information asymmetry, agents may pursue the maximisation of their own interests at the expense of the interests of the principal, resulting in so-called agency costs (Jensen & Meckling, 1976). The rise of social media helps to alleviate information asymmetry and expand the power of small investors, thus restricting the opportunistic behaviour of managers and improving corporate governance (Ang et al., 2021; Li & Zhang, 2023; Liu, 2022; Sun et al., 2022).

Specifically, social media can improve corporate governance in the following two ways. First, it can simultaneously improve the internal and external governance environment of the firm. From the perspective of the impact on internal governance, small investors' postings on social media can serve as supplementary information for major shareholders and the board of directors, which can enhance their cognition and supervision of the behaviour of managers and contribute to the overall improvement of internal governance. From the perspective of external governance, the sentiment from these social media postings generates a spillover effect, and its rapid spread is easily captured by the regulatory authorities, news media and auditors in the external market, thus facilitating the stricter external supervision of managers (Dyck et al., 2008; Liu, 2022; Sun et al., 2022). Second, postings on social media can alleviate the information asymmetry between small investors and listed firms. Information dissemination on social media has interactivity and real-time characteristics. It can immediately achieve interactive communication between investors and firms, as well as between investors and investors, promoting the flow of information (Cade, 2018) while also loosening the control firms have over their information environment (Miller & Skinner, 2015). This means that discussions from investors on social media increase the probability of managers' opportunistic behaviour being detected, thereby limiting their opportunistic motivation.

ESG greenwashing refers to the selective disclosure of ESG information to mislead investors and conceal poor ESG performance (Walker & Wan, 2012; Yu et al., 2020; Zhang, 2023a). Therefore, it is essentially an information manipulation behaviour based on opportunistic motivation. As mentioned earlier, social media platforms can improve the efficiency of information dissemination, thereby shortening the exposure time of negative information, while also increasing the risks for firms selectively disclosing environmental performance (Lyon & Montgomery, 2013). Although this information manipulation behaviour is difficult for small investors to detect in the short term, with the continuous accumulation of verifiable information, ESG

greenwashing is becoming more difficult (Jin & Myers, 2006). Pressure from both internal and external sources is unlikely to allow any inconsistency between corporate claims and actions to persist for a long time (Fiss & Zajac, 2006). Therefore, owing to the governance effect of social media, ESG greenwashing is inhibited.

Based on the above analysis, we propose Hypothesis H1b:

H1b Social media pressure inhibits firms from engaging in ESG greenwashing.

Research Design

Data and Sample

Our initial sample consists of all A-share listed firms from 2011 to 2021 because the limitations of Bloomberg ESG rating data. The sample data are processed as follows: (1) we delete the observation samples with missing values; (2) we exclude ST, *ST and PT trading status samples during the observation periods; and (3) we exclude samples of firms in the financial sector. Our final sample consists of 11,339 firm-year observations. In the data processing, all continuous variables are winsorised by 1%. In the regression process, we also conduct clustering adjustment at the firm level and use robust standard errors. The ESG information disclosure score data needed for the construction of an indicator of corporate ESG greenwashing are obtained from the Bloomberg terminal. The ESG performance score data are obtained from the China Securities ESG Rating, and the raw data are taken from the Flush iFinD financial data terminal.[5] We also consider the Wind ESG rating score as a robustness test. Unless otherwise specified, the other data in this paper come from China Stock Market & Accounting Research Database (CSMAR) and Chinese Research Data Services Platform (CNRDS).

[5] Following prior literature (Zhang, 2023a, 2023b), this paper uses the Bloomberg ESG information disclosure score to measure the level of ESG information disclosure, and the China Securities ESG rating to measure ESG performance. The Bloomberg ESG information disclosure score is only for the ESG information disclosure of listed firms, and changes in the range of [0, 100]. The higher the score, the better the ESG information disclosure status of the firm. Visit https://www.bloombergchina.com/solution/sustainable-finance/ for more information. The ESG rating of China Securities focuses on the actual ESG performance of Chinese enterprises, over nine grades (i.e. C, CC, CCC, B, BB, BBB, A, AA and AAA). According to the ESG rating of China Securities, we allocate actual ESG performance of listed firms from 1 to 9, respectively. Visit https://www.chindices.com/esg-data.html for more information.

Variable Construction and Model

Main Dependent Variable: ESG Greenwashing

Yu et al. (2020) create peer-relative ESG greenwashing scores for all sample firms by comparing the relative position of their ESG disclosure score with that for the ESG performance score. When a firm has a much better relative position than its peers in its ESG disclosure score than in its ESG performance score, this firm's greenwashing score will be positive, implying that it is likely obscuring its poor ESG performance by simply revealing large quantities of ESG data. Similarly, in this paper, the degree of ESG greenwashing is measured by the gap between the ESG disclosure score and the actual ESG performance score. Following existing research (Yu et al., 2020; Zhang, 2023a, 2023b), Model (1) is constructed to standardise the degree of ESG disclosure and actual ESG performance into the same scale by subtracting the mean value and dividing by the standard deviation, with the difference between the two used to measure the ESG greenwashing of the firm. That is, when a firm has a better position than its peers in the ESG disclosure score than in its ESG performance score, its ESG greenwashing score is positive:

$$Greenwashing_{i\,t} = \left[\frac{ESG_{Disclosure\,i,t} - \overline{ESG}_{Disclosure\,x}}{\sigma ESG_{Disclosure\,x}}\right] - \left[\frac{ESG_{Rating\,i,t} - \overline{ESG}_{Rating\,x}}{\sigma ESG_{Rating\,x}}\right] \quad (1)$$

where $ESG_{Disclosurei,t}$ and $ESG_{Ratingi,t}$ denote the ESG disclosure and ESG performance scores of firm i in year t, respectively; $\overline{ESG}_{Disclosurex}$ and $\overline{ESG}_{Ratingx}$ represent the means of the ESG disclosure and ESG performance scores in industry x;$\sigma_{ESG_{Disclosurex}}$ and $\sigma_{ESG_{Ratingx}}$ are the standard deviations of ESG disclosure and ESG performance scores, used to normalise the greenwashing scores. In our robustness tests, we replace the ESG performance scores with Wind ESG performance scores and remeasure ESG greenwashing.

Independent Variable: Negative Sentiment from Social Media

The *EastMoney* stock message board (*EastMoney*) comprises the majority of stock message board users in China. In addition, *EastMoney* provides a separate forum for each listed firm, and most of the participants in the forum are small investors who hold or are preparing to hold the firm's stocks, which helps us identify and measure a unique social media pressure for each listed firm. GUBA (Stock Comments Database) under CNRDS uses a machine learning

method to analyse the sentiment of the postings on the stock message boards, dividing these into three categories: positive, negative and neutral. We use Stata code to sum the daily data into annual data and calculate the negative sentiment index (*Neg-sent1*). This index not only considers the number of postings but also considers the overall sentiment of the stock message boards. In detail, we measure the independent variables based on the following Model (2):

$$Neg-sent1 = \frac{\text{negative postings per year} - \text{positive postings per year}}{\text{negative postings per year} + \text{positive postings per year}} * \ln(1 + \text{total postings per year}) \quad (2)$$

In addition, relative measure is the variable most commonly used in the textual analysis literature (Ang et al., 2021), and we also use the equation (negative postings per year + 1) / (positive postings per year + 1) to remeasure the negative sentiment of stock message boards *(Neg-sent2)*.

Control Variables

Following prior literature (Du et al., 2023; Li et al., 2018; Zhang, 2023a), the following control variables are selected for this study: the growth rate of revenue (*Growth*), firm size (*Size*), leverage ratio (*Lev*), ownership of executives (*Exeshare*), return on assets (*ROA*), whether the director or the chair is also the CEO (*Dual*), the ratio of independent board directors (*Indep*), board size (*Board*), ownership of the largest shareholders (*Top1*), state-owned enterprise status (*SOE*), firm age (*Age*) and whether the firm is in a heavily polluting industry (*Pollute*). The determination of heavily polluting industries refers to the 'Listed Companies' Environmental Protection Industry Classification Management Directory' issued by the Ministry of Environmental Protection of China in 2008, which classifies 16 industries, including thermal power, steel and cement, as polluting industries. In addition, we include the year fixed effect (*Year*) and the firm fixed effect (*Firm*). All control variables are defined in the Appendix.

Empirical Model

To examine whether negative sentiment from social media influences ESG greenwashing, we estimate the following Model (3):

$$Greenwashing_{it} = \beta_0 + \beta_1 Neg-sent1_{it}/Neg-sent2_{it} + \sum \beta\ Controls_{it} + Firm + Year + \varepsilon_{it} \quad (3)$$

where i and t represent firm i and year t, respectively. $Greenwashing_{it}$ is the proxy variable for ESG greenwashing as described above. The primary independent variables of interest are $Neg\text{-}sent1_{it}$ and $Neg\text{-}sent2_{it}$; if our Hypothesis H1a (H1b) is valid, we expect significantly positive (negative) coefficients (β1) on $Neg\text{-}sent1_{it}$ and $Neg\text{-}sent2_{it}$. $Controls_{it}$ is a set of firm-level variables as defined above, and ε_{it} is the error term. Finally, we include firm and year fixed effects to control for any time-invariant firm characteristics and time trends, respectively. Robust standard errors are clustered at the firm level to correct for firm-specific autocorrelation in the estimation errors.

Empirical Results

Descriptive Statistics

Table 1 reports the descriptive statistics of the key variables in this paper. Among them, the mean value of dependent variable *Greenwashing* is − 0.008, which illustrates that the average level of greenwashing in our sample is relatively low. However, similarly to the findings of Zhang (2023a), the maximum value of 5.432 and minimum value of − 3.509 indicate a large variation in the greenwashing behaviours of the sampled firms. The median of our key independent variable *Neg-sent1* (*Neg-sent2*) is − 1.160 (0.771), which illustrates that the number of negative postings is low relative to than the number of positive postings, indicating that the opinion from stock message boards shows more positive sentiment. The data characteristics of other control variables are relatively consistent with other studies, indicating that the selection of variables is reasonable.

Test of H1: Social Media Pressure and ESG Greenwashing

Table 2 presents the baseline regression results. Columns (1) and (3) report the regression results without firm-level control variables. The coefficients of the independent variables are 0.040 and 0.220, respectively, and are significant at the 1% level. After all the control variables are included, the results of Columns (2) and (4) show that the regression coefficients of the independent variables are still significant at the 1% level, indicating that negative sentiment on stock message boards significantly induces ESG greenwashing.

In terms of economic significance, in Column (2), a one-standard-deviation increase in the negative sentiment index (*Neg-sent1*) is associated with an increase of 3.94% (0.041 × 1.225/1.274) of a standard deviation in the degree of ESG greenwashing. In Column (4), a one-standard-deviation increase in the negative sentiment index (*Neg-sent2*) is associated with an increase of 4.06% (0.234 × 0.221/1.274) of a standard deviation in the degree of ESG greenwashing.

Table 1 Descriptive statistics

Variable	N	Mean	SD	Min	P25	Median	P75	Max
Greenwashing	10,886	− 0.008	1.274	− 3.509	− 0.938	− 0.074	0.816	5.432
Neg-sent1	10,886	− 1.218	1.225	− 4.475	− 1.975	− 1.160	− 0.397	1.747
Neg-sent2	10,886	0.784	0.221	0.317	0.633	0.771	0.917	1.489
Growth	10,886	0.178	0.380	− 0.493	− 0.004	0.117	0.271	2.343
Size	10,886	23.172	1.318	20.484	22.238	23.058	23.959	26.961
Lev	10,886	0.476	0.200	0.067	0.321	0.488	0.630	0.886
Exeshare	10,886	0.076	0.155	0.000	0.000	0.000	0.049	0.666
ROA	10,886	0.051	0.063	− 0.168	0.017	0.042	0.080	0.249
Dual	10,886	0.218	0.413	0.000	0.000	0.000	0.000	1.000
Indep	10,886	0.375	0.055	0.333	0.333	0.364	0.429	0.571
Board	10,886	2.176	0.202	1.609	2.079	2.197	2.197	2.708
Top1	10,886	0.369	0.164	0.034	0.241	0.354	0.486	0.900
SOE	10,886	0.481	0.500	0.000	0.000	0.000	1.000	1.000
Age	10,886	2.436	0.734	0.000	2.079	2.639	2.996	3.466
Pollute	10,886	0. 304	0.460	0.000	0.000	1.000	1.000	1.000

This table presents descriptive statistics of the variables used in this paper. Definitions are presented in the Appendix

Thus, the effect of social media pressure on ESG greenwashing is both statistically significant and economically meaningful, proving Hypothesis H1a.

The coefficients of the control variables are generally consistent with prior literature (Zhang, 2023c). Specifically, the coefficients on *Size* and *Age* are significantly negative, and the coefficients on *Lev* and *Growth* are significantly positive, indicating that the degree of ESG greenwashing is lower for large and mature firms but higher for firms with high leverage and high revenue growth ratios.

Robustness Tests

Alternative Measure of Dependent and Independent Variables

To ensure the reliability of baseline results, and avoid the possible influence of factors such as measurement errors, first, we use the ratio of annual negative postings to total annual postings (*Neg-sent3*) to measure negative sentiment on stock message boards. This measurement method indicates that we have considered the neutral postings. In addition, following Ang et al. (2021), we also use an absolute measure based on the natural logarithm of the annual number of negative postings (*Neg-sent4*). Second, we use alternative ESG greenwashing indicators to show our conclusion is reliable. Following Zhang (2023a), we remeasure the ESG performance of firms via the Wind ESG rating score, and use Model (2) to recalculate *Greenwashing-wind.* The regression results in Table 3 show that the coefficients of the negative sentiment on stock message boards are significantly positive, again proving that the baseline results are robust.

Distinguish ESG Performance Score and ESG Disclosure Score

Based on the theoretical analysis above, influenced by social media pressure, firms may achieve ESG greenwashing by improving the level of ESG disclosure to cover up actual poor ESG performance. Thus, to further demonstrate our conclusion, we regress ESG performance score (*Esgra*) and ESG disclosure score (*Esgds*) on independent variables (*Neg-sent1/ Neg-sent2)*, respectively, with the regression results shown in Table 4. Columns (1) and (2) show that the negative sentiment on stock message boards (*Neg-sent1/ Neg-sent2*) is significantly negatively correlated with ESG performance score (*Esgra*) at the 1% level. However, Columns (3) and (4) show the estimation results by using ESG disclosure information, and we find that the coefficient of the negative sentiment on stock message boards (*Neg-sent1/ Neg-sent2*) is significantly positive at the 5% level. The above results show that, to cope with social media pressure, firms may reduce their actual ESG performance while improving the level of ESG information disclosure. These findings help better explain how greenwashing behaviours arise and provide evidence that a different choice of greenwashing measurement method does not change the findings of our study.

Lagged Independent Variables

To alleviate the endogeneity problem, and take into account that ESG greenwashing may have a certain time lag, we lag independent variables (*Neg-sent1/ Neg-sent2*) in year *t–1* and *t–2*, respectively, instead of year t in Model (3). The

Table 2 Social media pressure and ESG greenwashing

Variable	(1) *Greenwashing*	(2) *Greenwashing*	(3) *Greenwashing*	(4) *Greenwashing*
Neg-sent1	0.040***	0.041***		
	(3.634)	(3.881)		
Neg-sent2			0.220***	0.234***
			(3.540)	(3.875)
Growth		0.056**		0.057**
		(2.285)		(2.314)
Size		− 0.085**		− 0.086**
		(− 2.278)		(− 2.315)
Lev		0.304**		0.304**
		(2.028)		(2.031)
Exeshare		− 0.512**		− 0.506**
		(− 2.218)		(− 2.194)
ROA		0.179		0.182
		(0.657)		(0.667)
Dual		0.050		0.050
		(1.213)		(1.221)
Indep		0.269		0.271
		(0.710)		(0.715)
Board		0.165		0.164
		(1.251)		(1.245)
Top1		0.133		0.139
		(0.605)		(0.632)
SOE		− 0.117		− 0.118
		(− 1.316)		(− 1.332)
Age		− 0.274***		− 0.275***
		(− 4.160)		(− 4.172)
Pollute		0.244*		0.244*
		(1.943)		(1.947)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,886	10,886	10,886	10,886
Adj.R2	0.388	0.394	0.388	0.394

This table presents the baseline results for the relationship between social media pressure and ESG greenwashing. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

results, shown in Table 5, indicate that the coefficients of our focus variables are still positive and significant, consistent with our baseline results.

Alleviate Omitted Variables

To further alleviate the endogeneity problem, we add to the baseline model several plausible candidates of time-variant omitted variables, including the firm's Baidu search index (*Baidu*), a dummy variable of corporate loss (*Loss*) and analyst coverage (*Analyst*). In addition, regional economic development level (*GDP*) and regional Internet development level (*IE*) are included. Finally, we also include the degree of ESG greenwashing in the previous period (*L. Greenwashing*). We add these additional control variables to Model (3) and re-estimate the regressions. The results are presented in Table 6, where we continue to observe a significantly positive coefficient on the main variables of interest, *Neg-sent1* and *Neg-sent2*.

Instrumental Variable Approach

To further address endogeneity concerns, we adopt a 2SLS regression with instrumental variables. We use two instrumental variables. First, following the work of Luo and Xiong (2021), we use the breadth of interaction channels between

Table 3 Alternative measure of dependent and independent variables

Variable	(1) *Greenwashing*	(2) *Greenwashing*	(3) *Greenwashing-wind*	(4) *Greenwashing-wind*
Neg-sent3	0.885***			
	(2.633)			
Neg-sent4		0.093***		
		(5.781)		
Neg-sent1			0.036**	
			(2.131)	
Neg-sent2				0.200**
				(2.184)
Growth	0.055**	0.058**	0.074*	0.074*
	(2.226)	(2.358)	(1.932)	(1.941)
Size	− 0.086**	− 0.109***	0.048	0.048
	(− 2.301)	(− 2.926)	(0.714)	(0.710)
Lev	0.303**	0.293**	0.183	0.180
	(2.017)	(1.977)	(0.737)	(0.725)
Exeshare	− 0.520**	− 0.432*	− 0.624*	− 0.624*
	(− 2.255)	(− 1.888)	(− 1.758)	(− 1.762)
ROA	0.137	0.089	0.153	0.151
	(0.501)	(0.326)	(0.507)	(0.500)
Dual	0.048	0.050	0.075	0.076
	(1.170)	(1.216)	(1.323)	(1.333)
Indep	0.265	0.266	− 0.941	− 0.944
	(0.697)	(0.698)	(− 1.581)	(− 1.586)
Board	0.161	0.166	− 0.099	− 0.098
	(1.213)	(1.264)	(− 0.481)	(− 0.475)
Top1	0.117	0.224	− 0.728**	− 0.726**
	(0.533)	(1.022)	(− 2.077)	(− 2.070)
SOE	− 0.119	− 0.117	0.194	0.194
	(− 1.332)	(− 1.341)	(1.376)	(1.373)
Age	− 0.282***	− 0.272***	− 0.372*	− 0.373*
	(− 4.278)	(− 4.135)	(− 1.842)	(− 1.837)
Pollute	0.243*	0.252**	0.129	0.128
	(1.937)	(2.025)	(0.660)	(0.657)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,886	10,886	4374	4374
Adj.R2	0.393	0.396	0.010	0.010

This table presents the regression results of alternative measures of dependent and independent variables. *Neg-sent3* is the ratio of annual negative postings to total annual postings. *Neg-sent4* is the natural logarithm of the annual number of negative postings. *Greenwashing-wind* is calculated via the Wind ESG rating score. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the Appendix

firms and investors (*Interaction*) as the first instrumental variable. When there is an investor relationship management or investor consultation section on the official website homepage of a firm, or an official interactive channel on WeChat and Weibo, the value of *Interaction* is 1; otherwise, it is 0. These data are sourced from CNRDS database. The rationale is that more investor–firm interaction platforms represent more communication options for small investors; therefore, the number of postings on *EastMoney* will be relatively fewer. Moreover, if a firm attaches importance to daily investor relationship management, the sentiment from social media is likely to be less negative; however, the establishment of such platforms are unlikely to directly affect ESG greenwashing. Second, we use the natural logarithm of the

Table 4 Distinguish ESG performance score and ESG disclosure score

Variable	(1) *Esgra*	(2) *Esgra*	(3) *Esgds*	(4) *Esgds*
Neg-sent1	− 0.036***		0.142**	
	(− 3.244)		(2.364)	
Neg-sent2		− 0.212***		0.735**
		(− 3.323)		(2.218)
Growth	− 0.089***	− 0.090***	− 0.008	− 0.008
	(− 3.564)	(− 3.600)	(− 0.069)	(− 0.063)
Size	0.279***	0.280***	1.227***	1.224***
	(7.409)	(7.438)	(5.620)	(5.611)
Lev	− 0.736***	− 0.737***	− 3.107***	− 3.109***
	(− 4.643)	(− 4.649)	(− 4.241)	(− 4.243)
Exeshare	0.916***	0.909***	2.664**	2.673**
	(3.389)	(3.365)	(2.238)	(2.247)
ROA	0.275	0.269	4.392***	4.374***
	(0.931)	(0.911)	(3.299)	(3.287)
Dual	− 0.058	− 0.059	0.009	0.009
	(− 1.440)	(− 1.449)	(0.039)	(0.038)
Indep	0.151	0.149	4.406**	4.409**
	(0.390)	(0.385)	(2.273)	(2.274)
Board	− 0.083	− 0.083	1.233*	1.228*
	(− 0.624)	(− 0.620)	(1.901)	(1.894)
Top1	0.158	0.152	2.360*	2.368*
	(0.663)	(0.637)	(1.940)	(1.946)
SOE	0.095	0.096	− 0.416	− 0.419
	(0.972)	(0.986)	(− 0.839)	(− 0.846)
Age	0.438***	0.439***	1.179***	1.173***
	(6.582)	(6.587)	(3.015)	(2.997)
Pollute	0.030	0.029	0.111	0.111
	(0.247)	(0.245)	(0.223)	(0.223)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,886	10,886	10,886	10,886
Adj.R2	0.058	0.058	0.693	0.693

This table presents the regression results of distinguishing ESG performance score and ESG disclosure score. *Esgra* is the ESG performance score and *Esgds* is the ESG disclosure score. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

number of Internet broadband access users at the province level (*BroadBand*) as the second instrumental variable. The rationale is that one prerequisite for small investors to speak out on social media is a well-functioning Internet infrastructure. If a province has more Internet broadband users, more small investors in the province are likely to communicate on social media, including on stock message boards. Huang et al. (2016) find that small investors pay more attention to stocks of local firms than to those of nonlocal firms. Therefore, firms in provinces with more Internet broadband access users will receive more social media attention. In particular, based on negative bias theory in the field of psychology, negative sentiment on social media is more likely to spread among investors (Li et al., 2020), so these firms may also face more pressure from social media; however, the level of Internet development at the regional level is unlikely to directly affect a firm's greenwashing behaviour.

Table 7 reports the results of the instrumental variable approach. In the first-stage estimations in Columns (1) and (3), the coefficients of *Interaction* are significantly negative and the coefficients of *BroadBand* are significantly positive, indicating that the instrumental variable satisfies the relevance criterion. Further, the Kleibergen-Paap rk Wald F statistics are 58.384 and 58.020 in Columns (1) and (3), respectively, and thus we reject the weak instrumental variable hypothesis. Additionally, Hansen's J-test in the second stage suggests that we cannot reject the null hypothesis that the instrumental variables are not associated with the second stage's residual term. Therefore, the statistics indicate that the instrumental variables are exogenous and effective. The second-stage regression results in Columns (2) and (4) show that the coefficients of negative sentiment on social media (*Neg-sent1/Neg-sent2*) are still significantly positive at the 5% level, which is consistent with the baseline results. Thus, our results are unchanged.

Heckman Two-stage Model

In this section, we use the Heckman two-stage model to alleviate concerns about selection bias. The disclosure of an ESG report is voluntary and ESG rating agencies such as Bloomberg also have limited coverage of Chinese listed firms. Therefore, the results of this paper may be affected by significant sample selection bias. To address this concern, following prior literature (Du et al., 2023), we adopt a Heckman two-step sample selection model. In the first stage of the Heckman two-stage model, we set the dummy variable (*Greenwashing_Dum*) as 1 if *Greenwashing* is not a missing value, and as 0 otherwise. Then, we employ the full sample to conduct a probit regression and explore which factors may affect the possibility. The control variables we use are the same as for Model (3). Results of the analysis are presented in Column (1) of Table 8. Next, we calculate an inverse Mills' ratio (*IMR*) using the estimates from the first stage and add this as the control variable in the second-stage regression.

In the second stage, we examine whether social media pressure still induces ESG greenwashing after adding the inverse Mills' ratio as a control variable. Results of the analysis are presented in Columns (2) and (3) of Table 8. We continue to observe a significantly positive coefficient

Table 5 Lagged independent variables

Variable	(1) *Greenwashing*	(2) *Greenwashing*	(3) *Greenwashing*	(4) *Greenwashing*
L.Neg-sent1	0.018*			
	(1.865)			
L2.Neg-sent1		0.018*		
		(1.730)		
L.Neg-sent2			0.126**	
			(2.348)	
L2.Neg-sent2				0.130**
				(2.227)
Growth	0.045*	0.024	0.045*	0.024
	(1.798)	(0.876)	(1.819)	(0.880)
Size	− 0.073***	− 0.084***	− 0.073***	− 0.084***
	(− 2.742)	(− 2.679)	(− 2.761)	(− 2.690)
Lev	0.291***	0.262**	0.291***	0.264**
	(2.628)	(2.093)	(2.632)	(2.111)
Exeshare	− 0.612***	− 0.537**	− 0.605***	− 0.531**
	(− 3.334)	(− 2.523)	(− 3.294)	(− 2.491)
ROA	− 0.021	− 0.169	− 0.011	− 0.165
	(− 0.100)	(− 0.742)	(− 0.053)	(− 0.724)
Dual	0.052*	0.056*	0.052*	0.056*
	(1.653)	(1.655)	(1.673)	(1.655)
Indep	0.020	0.029	0.023	0.029
	(0.069)	(0.091)	(0.080)	(0.091)
Board	0.123	0.192*	0.122	0.191*
	(1.281)	(1.815)	(1.276)	(1.806)
Top1	0.044	− 0.178	0.048	− 0.174
	(0.275)	(− 0.940)	(0.297)	(− 0.921)
SOE	− 0.163**	− 0.114	− 0.165**	− 0.115
	(− 2.440)	(− 1.545)	(− 2.463)	(− 1.562)
Age	− 0.176***	− 0.017	− 0.176***	− 0.018
	(− 2.641)	(− 0.173)	(− 2.638)	(− 0.186)
Pollute	0.315***	0.468***	0.316***	0.468***
	(3.682)	(4.798)	(3.687)	(4.798)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	9257	7920	9257	7920
Adj.R2	0.312	0.293	0.312	0.293

This table presents the regression results of lagging independent variables in year t–1 and t–2. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

of the main variables of interest, *Neg-sent1* and *Neg-sent2*. Moreover, the inverse Mills' ratio (*IMR*) is significant in the second-stage regression, suggesting that selection bias does present in our analysis. To sum up, the association between social media pressure and ESG greenwashing is still positive, as evidenced by the two-stage Heckman model.

Additional Tests

In this section, to provide robust evidence that it is indeed the social media pressure that induces ESG greenwashing, we conduct two additional tests.

Table 6 Alleviate omitted variables

Variable	(1) *Greenwashing*	(2) *Greenwashing*
Neg-sent1	0.028***	
	(2.655)	
Neg-sent2		0.159***
		(2.661)
Growth	0.042	0.042*
	(1.630)	(1.648)
Size	− 0.016	− 0.017
	(− 0.443)	(− 0.471)
Lev	0.161	0.162
	(1.171)	(1.177)
Exeshare	− 0.555**	− 0.554**
	(− 2.478)	(− 2.476)
ROA	− 0.015	− 0.015
	(− 0.047)	(− 0.048)
Dual	0.025	0.025
	(0.706)	(0.715)
Indep	− 0.102	− 0.102
	(− 0.320)	(− 0.319)
Board	0.025	0.024
	(0.215)	(0.211)
Top1	− 0.083	− 0.079
	(− 0.403)	(− 0.385)
SOE	− 0.131	− 0.132
	(− 1.417)	(− 1.423)
Age	− 0.116	− 0.117
	(− 1.385)	(− 1.397)
Pollute	0.085	0.085
	(0.515)	(0.519)
Baidu	0.000**	0.000**
	(2.066)	(2.035)
Loss	0.070*	0.069*
	(1.770)	(1.754)
Analyst	− 0.015	− 0.015
	(− 1.017)	(− 1.023)
GDP	− 0.468***	− 0.470***
	(− 2.916)	(− 2.928)
IE	0.000	0.000
	(0.155)	(0.143)
L. Greenwashing	0.357***	0.357***
	(25.618)	(25.616)
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	8283	8283
Adj.R2	0.499	0.499

This table presents the regression results controlling for possibly omitted variables. In addition to the control variables in Table 2, we include the firm's Baidu search index (*Baidu*), a dummy variable of corporate loss (*Loss*), analyst coverage (*Analyst*), regional economic development level (*GDP*), regional Internet development level (*IE*) and the degree of ESG greenwashing in the previous period (*L. Greenwashing*). Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

Stock Message Board Attention and Social Media Pressure

Here, we examine whether the effect of social media pressure on ESG greenwashing varies with stock message board attention. We argue that the impact of small investors relies on stock message boards' function of enhancing their social influence. If this is the case, the effect should be stronger under greater attention of stock message boards. This is because the more influential stock message boards become, the higher the probability of small investors' postings being browsed and thence generating a 'public opinion storm'. In these circumstances, managers should pay more attention to opinions on social media and be more likely to conduct ESG greenwashing.

To test our conjecture, we divide the sample into two groups based on stock message board attention (*Readratio*): a high-attention group, when the browsing proportion of postings (read volume of postings per year / number of total postings per year) is above the industry-year median, and vice versa for a low-attention group. We re-estimate Model (3) based on the two subsamples and report the results in Table 9. The coefficients of *Neg-sent1* (*Neg-sent2*) are significantly positive in Columns (2) and (4) at the 1% level, but insignificant in Columns (1) and (3). These results support the notion that it is stock message board discussion that empowers small investors to exert influence on managerial decision-making.

Executives' Age and Social Media Pressure

Here, we examine whether the effect of social media pressure on ESG greenwashing varies with executives' age. In fact, an important premise for small investors' effect is that executives are willing to access and consider online information. Thus, it is reasonable to infer that this mechanism is more powerful when executives have more exposure to online information. Choi et al. (2002) find that young, male and wealthy participants are more likely to try web channels. According to the 45th Statistical Report on the Development of the Internet in China published by China Internet Information Center (CNNIC) in 2020, 61.6% of Internet users in China are 10–39 years old. Among groups over 20, Internet use decreases with age. Consequently, following Sun et al. (2022), we estimate whether the effect of interest is stronger under circumstances where executives are younger and thus more heavily exposed to the impact of Internet information.

To test our conjecture, we calculate *Exeage* as the average age of a company's executives and divide the sample into two groups based on the industry-year median of *Exeage*. We re-estimate Model (3) based on the two subsamples and report the results in Table 10. The coefficients

Table 7 Instrumental variable approach

Variable	(1) *Neg-sent1*	(2) *Greenwashing*	(3) *Neg-sent2*	(4) *Greenwashing*
Interaction	− 0.011**		− 0.002**	
	(− 1.980)		(− 1.997)	
BroadBand	0.023***		0.004***	
	(10.779)		(10.741)	
Neg-sent1		0.120**		
		(1.977)		
Neg-sent2				0.642**
				(1.989)
Growth	− 0.083***	0.061**	− 0.017***	0.062**
	(− 2.648)	(2.266)	(− 3.026)	(2.292)
Size	0.004	− 0.118***	0.009	− 0.123***
	(0.077)	(− 2.596)	(1.017)	(− 2.700)
Lev	− 0.130	0.290*	− 0.030	0.293*
	(− 0.742)	(1.652)	(− 0.905)	(1.670)
Exeshare	− 0.729**	− 0.625**	− 0.139**	− 0.623**
	(− 2.443)	(− 2.208)	(− 2.556)	(− 2.213)
ROA	− 2.661***	0.354	− 0.480***	0.341
	(− 9.164)	(1.009)	(− 9.125)	(0.979)
Dual	− 0.073*	0.036	− 0.015*	0.037
	(− 1.702)	(0.728)	(− 1.934)	(0.751)
Indep	− 0.037	0.062	− 0.002	0.059
	(− 0.103)	(0.143)	(− 0.035)	(0.137)
Board	− 0.116	0.143	− 0.016	0.139
	(− 0.831)	(0.899)	(− 0.655)	(0.877)
Top1	− 0.676***	0.205	− 0.142***	0.214
	(− 2.865)	(0.867)	(− 3.318)	(0.909)
SOE	0.024	− 0.191*	0.010	− 0.195*
	(0.253)	(− 1.797)	(0.584)	(− 1.840)
Age	− 0.340***	− 0.307***	− 0.055***	− 0.312***
	(− 3.919)	(− 3.857)	(− 3.480)	(− 3.964)
Pollute	− 0.127	0.166	− 0.025	0.167
	(− 0.992)	(1.112)	(− 1.070)	(1.119)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F statistic	58. 384		58.020	
p-value for Hansen J statistic		0.512		0.511
N	8377	8377	8377	8377
Adj.R2	0.567	0.000	0.548	0.001

This table presents the 2SLS results for the relationship between social media pressure and ESG greenwashing. *Interaction* is the first instrumental variable. When there is an investor relationship management or investor consultation section on the official website homepage of firm, or an official interactive channel on WeChat and Weibo, the value of *Interaction* is 1; otherwise, it is 0. *BroadBand* is the second instrumental variable, defined as the natural logarithm of the number of Internet broadband access users at the province level. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

of *Neg-sent1* (*Neg-sent2*) are significantly positive in Columns (2) and (4) at the 1% level, but less significant in Columns (1) and (3). The results imply that, in groups with older managers, social media pressure has a lower effect on ESG greenwashing. This indicates a weaker small investor influence when executives have limited subjective

Table 8 Heckman two-stage model

	(1) First step	(2) Second step	(3)
	Greenwashing _dum	*Greenwashing*	*Greenwashing*
Neg-sent1		0.025**	
		(2.034)	
Neg-sent2			0.156**
			(2.324)
Growth	− 0.012***	− 0.065	− 0.064
	(− 3.728)	(− 0.871)	(− 0.860)
Size	0.105***	1.144*	1.141*
	(13.843)	(1.851)	(1.847)
Lev	− 0.135***	− 1.179	− 1.176
	(− 6.121)	(− 1.471)	(− 1.468)
Exeshare	0.025	− 0.050	− 0.044
	(0.587)	(− 0.169)	(− 0.147)
ROA	0.032	0.110	0.117
	(0.968)	(0.333)	(0.355)
Dual	0.011*	0.170**	0.170**
	(1.792)	(2.050)	(2.054)
Indep	0.043	− 0.959*	− 0.958*
	(0.604)	(− 1.944)	(− 1.944)
Board	0.015	0.274	0.274
	(0.630)	(1.579)	(1.578)
Top1	0.069	0.990**	0.995**
	(1.408)	(2.191)	(2.202)
SOE	0.007	− 0.160	− 0.161
	(0.534)	(− 1.496)	(− 1.505)
Age	− 0.010	− 0.006	− 0.005
	(− 1.268)	(− 0.065)	(− 0.058)
Pollute	− 0.005	− 0.166	− 0.166
	(− 0.241)	(− 1.250)	(− 1.252)
IMR		21.643**	21.610**
		(2.039)	(2.037)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	33,619	10,931	10,931
Adj.R2	0.369	0.011	0.011

This table presents the regression results of Heckman's two-step model. Column (1) reports the results of the first-step regression, and Columns (2) and (3) report the results of the second-step regressions. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

initiative to use social media, consistent with the findings of Sun et al. (2022).

Table 9 Stock message board attention and social media pressure

Variable	*Greenwashing*		*Greenwashing*	
	(1) Attention-low	(2) Attention-high	(3) Attention-low	(4) Attention-high
Neg-sent1	0.024	0.044***		
	(1.533)	(3.009)		
Neg-sent2			0.101	0.258***
			(1.108)	(3.238)
Growth	0.068**	0.051	0.067**	0.052
	(2.000)	(1.333)	(1.969)	(1.371)
Size	− 0.093	− 0.105**	− 0.092	− 0.107**
	(− 1.582)	(− 2.164)	(− 1.573)	(− 2.197)
Lev	0.426**	0.055	0.424**	0.057
	(1.989)	(0.276)	(1.977)	(0.283)
Exeshare	− 0.548	− 0.341	− 0.553	− 0.332
	(− 1.523)	(− 1.093)	(− 1.541)	(− 1.064)
ROA	− 0.013	− 0.110	− 0.025	− 0.102
	(− 0.034)	(− 0.285)	(− 0.064)	(− 0.265)
Dual	0.013	0.107**	0.012	0.108**
	(0.212)	(2.013)	(0.204)	(2.021)
Indep	− 0.877*	1.037**	− 0.875*	1.036**
	(− 1.730)	(2.144)	(− 1.726)	(2.143)
Board	− 0.017	0.205	− 0.019	0.203
	(− 0.102)	(1.161)	(− 0.110)	(1.149)
Top1	0.211	0.096	0.211	0.108
	(0.707)	(0.319)	(0.704)	(0.358)
SOE	− 0.167	− 0.087	− 0.167	− 0.090
	(− 1.444)	(− 0.550)	(− 1.444)	(− 0.571)
Age	− 0.248***	− 0.343***	− 0.250***	− 0.343***
	(− 2.656)	(− 3.832)	(− 2.684)	(− 3.825)
Pollute	0.270	0.229	0.270	0.229
	(1.275)	(1.380)	(1.277)	(1.385)
Diff	0.052*		0.028**	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	5361	5525	5361	5525
Adj.R2	0.411	0.361	0.411	0.361

This table estimates whether small investors' influence is stronger under greater attention on stock message boards. *Readratio* is the browsing proportion of postings. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

Channel Tests

Based on the theoretical analysis of this paper, social media pressure may strengthen managers' motivation for impression management and induce the opportunistic behaviour of ESG greenwashing. The underlying premise of this logic is that managers are myopic owing to the pressure for

short-term performance, and thus attempt to conduct ESG greenwashing rather than ESG practices. As such, we predict that social media pressure should have a positive effect on managerial myopia, which further leads to ESG greenwashing. Following Chen et al. (2024), to test the prediction, we estimate the following two-stage regressions:

$$M_{it} = \beta_0 + \beta_1 Neg - sent_{it} + \sum \beta Controls_{it} + Firm + Year + \varepsilon_{it} \quad (4)$$

$$Greenwashing_{it} = \beta_0 + \beta_1 \hat{M}_{it} + \sum \beta Controls_{it} + Firm + Year + \varepsilon_{it} \quad (5)$$

where M in Model (4) is our mechanism analysis variable, namely, managerial myopia. In the first stage, we regress managerial myopia on social media pressure and the set of control variables in our baseline model. In the second stage, the key independent variable $\widehat{M}$ is the predicted managerial myopia based on the first-stage regression (4), which captures the part of managerial myopia that can be explained by social media pressure. If managerial myopia is the possible mechanism through which social media pressure affects ESG greenwashing, the coefficients on M and $\widehat{M}$ should be positive and significant.

Specifically, we measure managerial myopia in two ways. First, Lai et al. (2020) suggest that a reduction in corporate R&D expenses can indicate managerial myopia. On the one hand, most R&D expenses reported as period expenses directly reduce the current year's revenue, which is not conducive to the short-term performance of firms. On the other hand, R&D expenditure, which is long term and risky, is difficult for external investors to detect. Therefore, myopic managers are more motivated to surreptitiously reduce R&D expenses. Following Chen et al. (2015) and Lai et al. (2020), we define *CutRD* as an indicator variable that equals 1 if year *t*'s R&D expense is lower than that of year *t-1*, and 0 otherwise. As shown in Columns (1) and (3) of Table 11, the coefficients on the negative sentiment of stock message boards *(Neg-sent1/Neg-sent2)* are all positive and significant. This indicates that social media pressure can exacerbate managerial myopia. Columns (2) and (4) present the second-stage regression. The key independent variable, $\widehat{\text{CutRD}}$, is the predicted *CutRD* based on the first-stage regression. We find that the coefficients on $\widehat{\text{CutRD}}$ are all positive and significant. These results are consistent with our prediction that managerial myopia is associated with a high level of ESG greenwashing.

Second, we use text analysis to measure managerial myopia based on the myopic behaviour words proposed by Brochet et al. (2015). Language can reflect people's cognition, preferences and personality, and researchers can capture these characteristics by analysing the type and frequency of words used in the language of experimental subjects (Pennebaker et al., 2003). Manager discussion and analysis (MD&A) as disclosed in the annual reports of listed firms is managers' review of the operating conditions of the firm during the reporting period and the prospect of its future development. Using MD&A to capture the subconscious cognition and characteristics of managers is a widely recognised research method (Li, 2010). Specifically, following Zhang et al. (2023), we measure managerial myopia (*Index*) by dividing the number of myopic behaviour words by the total number of MD&A words and multiplying it by 100. A higher proportion of myopic behavioural words indicates a more serious level of managerial myopia. As shown in Columns (1) and (3) of Table 12, the coefficients on the negative sentiment of stock message boards *(Neg-sent1/Neg-sent2)* are all positive and significant, indicating that social media pressure is positively related to managerial myopia. In addition, Columns (2) and (4) show that the coefficients on $\widehat{\text{Index}}$ are all positive and significant at the 1% level, suggesting that managerial myopia is associated with a high level of ESG greenwashing. In sum, the results in Tables 11 and 12 provide robust evidence that increased managerial myopia is an important channel through which social media pressure affects ESG greenwashing.

Cross-sectional Tests

The above theoretical analysis and empirical results show that social media pressure can exacerbate managerial myopia and induce firms to conduct impression management through ESG greenwashing. In this section, we conduct cross-sectional tests from dimensions of market value and reputation to further confirm the impression management motivation of ESG greenwashing under pressure from social media.

Market Value

Market value is an important indicator for measuring the strength of a firm and is also the key to ensuring the effectiveness of management equity incentives. As corporate insiders, managers have the motivation and ability to manage market value through opportunistic behaviours and maximise their personal interest (Chan et al., 2010). We believe that firms with lower market value come under greater pressure from social media, and so are more motivated to conduct impression management through ESG greenwashing.

In this paper, we partition our sample based on the median value of market value (*TobinQ*) in an industry year and regress for subsamples with high vs. low levels of market value. Table 13 shows the results of the cross-sectional analysis of market value. The results displayed in Columns (1) and (3) show that in the group with low market value, the negative sentiment of stock message boards (*Neg-sent1/*

Table 10 Executives' age and social media pressure

Variable	*Greenwashing*		*Greenwashing*	
	(1) Executives' age-low	(2) Executives' age-high	(3) Executives' age-low	(4) Executives' age-high
Neg-sent1	0.050***	0.037*		
	(6.389)	(1.784)		
Neg-sent2			0.282***	0.204
			(6.390)	(1.619)
Growth	0.011	0.117***	0.012	0.117***
	(0.405)	(4.415)	(0.451)	(4.363)
Size	− 0.096***	− 0.140***	− 0.098***	− 0.141***
	(− 4.483)	(− 3.831)	(− 4.570)	(− 3.839)
Lev	0.291	0.098	0.290	0.100
	(1.692)	(0.444)	(1.689)	(0.447)
Exeshare	− 0.369	− 0.196	− 0.359	− 0.196
	(− 1.668)	(− 0.643)	(− 1.616)	(− 0.648)
ROA	0.133	0.447	0.136	0.446
	(0.728)	(1.386)	(0.737)	(1.385)
Dual	0.054	0.040	0.054	0.040
	(0.655)	(0.724)	(0.648)	(0.728)
Indep	0.255	0.524	0.257	0.526
	(0.320)	(1.273)	(0.324)	(1.271)
Board	0.243	0.224	0.241	0.224
	(0.710)	(1.620)	(0.704)	(1.616)
Top1	0.009	0.489**	0.013	0.494**
	(0.035)	(2.096)	(0.048)	(2.093)
SOE	− 0.170	− 0.054	− 0.173	− 0.055
	(− 1.385)	(− 0.565)	(− 1.409)	(− 0.577)
Age	− 0.192	0.038	− 0.200	0.039
	(− 0.396)	(0.082)	(− 0.413)	(0.086)
Pollute	0.362**	− 0.055	0.364**	− 0.057
	(2.589)	(− 0.362)	(2.580)	(− 0.369)
Diff	0.062*		0.079*	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	5335	5300	5335	5300
Adj.R2	0.663	0.685	0.663	0.685

This table estimates whether small investors' effect is stronger under circumstances where executives are more heavily exposed to the effect of social media. *Exeage* is the average age of a company's executives. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

Neg-sent2) is significantly positive at the 1% level. However, as shown in Columns (2) and (4), in the group with high market value, the coefficients of the main independent variables are not statistically significant. This means that, under pressure from social media, the impression management motivation for ESG greenwashing is stronger in listed firms with lower market value.

Reputation

The occurrence of negative events severely affects firm reputation, while positive social responsibility activities and information disclosure have a 'reputation insurance' effect that can help firms reduce their losses following negative events (Anantharaman et al., 2022). Therefore, firms generally take the initiative to improve the disclosure level of social responsibility information after suffering a negative shock (Deegan, 2002). Thus, facing pressure from social

Table 11 Channel tests

Variable	(1) *CutRD*	(2) *Greenwashing*	(3) *CutRD*	(4) *Greenwashing*
Neg-sent1	0.038**			
	(2.545)			
$\widehat{\text{CutRD1}}$		0.673**		
		(2.266)		
Neg-sent2			0.217***	
			(2.674)	
$\widehat{\text{CutRD2}}$				0.691**
				(2.329)
Growth	− 0.725***	0.139***	− 0.724***	0.142***
	(− 9.350)	(3.009)	(− 9.340)	(3.065)
Size	0.003	− 0.090**	0.001	− 0.090**
	(0.158)	(− 2.427)	(0.083)	(− 2.425)
Lev	− 0.238**	0.349**	− 0.236**	0.350**
	(− 2.039)	(2.315)	(− 2.025)	(2.321)
Exeshare	− 0.251**	− 0.402*	− 0.249**	− 0.400*
	(− 2.052)	(− 1.752)	(− 2.041)	(− 1.744)
ROA	− 3.584***	0.875**	− 3.581***	0.897**
	(− 10.769)	(2.080)	(− 10.765)	(2.129)
Dual	− 0.024	0.049	− 0.025	0.050
	(− 0.589)	(1.199)	(− 0.603)	(1.205)
Indep	− 0.001	0.262	− 0.000	0.262
	(− 0.002)	(0.684)	(− 0.000)	(0.684)
Board	− 0.078	0.161	− 0.079	0.161
	(− 0.793)	(1.200)	(− 0.803)	(1.202)
Top1	0.279**	0.091	0.282**	0.091
	(2.390)	(0.409)	(2.411)	(0.406)
SOE	− 0.034	− 0.111	− 0.034	− 0.111
	(− 0.798)	(− 1.231)	(− 0.799)	(− 1.231)
Age	0.030	− 0.196**	0.029	− 0.196**
	(1.012)	(− 2.448)	(0.994)	(− 2.450)
Pollute	− 0.108**	0.303**	− 0.111***	0.305**
	(− 2.557)	(2.414)	(− 2.619)	(2.428)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,774	10,774	10,774	10,774
Adj.R2	0.089	0.398	0.089	0.398

This table presents the two-stage regression results of channel tests. *CutRD* is an indicator variable that equals 1 if year t's R&D expense is lower than that of year t-1, and 0 otherwise. $\widehat{\text{CutRD}}$ is the predicted *CutRD* based on the first-stage regression. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

Table 12 Channel tests

Variable	(1) *Index*	(2) *Greenwashing*	(3) *Index*	(4) *Greenwashing*
Neg-sent1	0.003*			
	(1.761)			
$\widehat{\text{Index1}}$		17.962***		
		(4.157)		
Neg-sent2			0.015*	
			(1.768)	
$\widehat{\text{Index2}}$				18.603***
				(4.224)
Growth	− 0.008**	0.198***	− 0.008**	0.204***
	(− 2.494)	(4.535)	(− 2.482)	(4.611)
Size	− 0.011*	0.122**	− 0.011*	0.128**
	(− 1.952)	(2.088)	(− 1.958)	(2.178)
Lev	0.047**	− 0.586**	0.047**	− 0.617**
	(2.401)	(− 2.281)	(2.395)	(− 2.354)
Exeshare	− 0.054*	0.458	− 0.054*	0.494
	(− 1.771)	(1.369)	(− 1.767)	(1.475)
ROA	− 0.086**	1.489***	− 0.086**	1.547***
	(− 2.350)	(3.177)	(− 2.352)	(3.248)
Dual	− 0.007	0.166***	− 0.007	0.170***
	(− 1.259)	(3.198)	(− 1.258)	(3.244)
Indep	− 0.080	1.490***	− 0.080	1.542***
	(− 1.522)	(2.766)	(− 1.520)	(2.862)
Board	− 0.020	0.489***	− 0.020	0.503***
	(− 1.125)	(2.905)	(− 1.127)	(2.983)
Top1	0.014	− 0.220	0.014	− 0.227
	(0.449)	(− 0.841)	(0.460)	(− 0.870)
SOE	− 0.020	0.259*	− 0.020	0.271**
	(− 1.332)	(1.910)	(− 1.336)	(1.981)
Age	0.007	− 0.311***	0.007	− 0.314***
	(0.791)	(− 3.735)	(0.784)	(− 3.779)
Pollute	− 0.000	0.392**	− 0.000	0.392**
	(− 0.012)	(2.321)	(− 0.013)	(2.325)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	9275	9275	9275	9275
Adj.R2	0.032	0.385	0.032	0.385

This table presents the two-stage regression results of channel tests. *Index* is total myopic behaviour words × 100/total MD&A words. $\widehat{\text{Index}}$ is the predicted *Index* based on the first-stage regression. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

media, we suggest that firms with poor reputation (more negative events) are more likely to repair their reputation through ESG greenwashing.

To test our conjecture, we partition our sample based on the median value of the number of negative media coverage on listed firms in an industry year and regress for subsamples with high vs. low levels of corporate reputation. Specifically, we collect data from CNRDS 'Financial News Database of Chinese Listed Companies' on the total number of annual negative media coverage. Table 14 shows the results of the

Table 13 Cross-sectional tests of market value

Variable	*Greenwashing*		*Greenwashing*	
	(1) Market value-low	(2) Market value-high	(3) Market value-low	(4) Market value-high
Neg-sent1	0.076***	0.014		
	(4.861)	(0.950)		
Neg-sent2			0.441***	0.071
			(4.943)	(0.838)
Growth	0.069*	0.066*	0.071*	0.066*
	(1.817)	(1.954)	(1.857)	(1.953)
Size	− 0.121**	− 0.065	− 0.123**	− 0.065
	(− 2.080)	(− 1.201)	(− 2.126)	(− 1.204)
Lev	0.630***	0.186	0.632***	0.185
	(2.628)	(0.985)	(2.635)	(0.981)
Exeshare	− 0.779	− 0.187	− 0.777	− 0.185
	(− 1.317)	(− 0.816)	(− 1.317)	(− 0.806)
ROA	− 0.029	0.073	− 0.026	0.071
	(− 0.057)	(0.220)	(− 0.051)	(0.212)
Dual	0.024	0.085	0.024	0.085
	(0.451)	(1.414)	(0.439)	(1.414)
Indep	0.747	− 0.113	0.756	− 0.111
	(1.615)	(− 0.188)	(1.639)	(− 0.185)
Board	0.356*	0.024	0.358*	0.024
	(1.893)	(0.122)	(1.905)	(0.122)
Top1	− 0.182	− 0.032	− 0.168	− 0.031
	(− 0.591)	(− 0.096)	(− 0.547)	(− 0.094)
SOE	0.111	− 0.212**	0.108	− 0.212**
	(0.751)	(− 1.994)	(0.735)	(− 1.997)
Age	− 0.361***	− 0.247***	− 0.360***	− 0.248***
	(− 3.439)	(− 2.636)	(− 3.436)	(− 2.648)
Pollute	0.310**	0.129	0.306**	0.129
	(2.180)	(0.738)	(2.164)	(0.740)
Diff	0.006***		0.019**	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	5423	5463	5423	5463
Adj.R2	0.032	0.385	0.032	0.385

This table presents the cross-sectional tests of market value. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

heterogeneity analysis of corporate reputation. The results in Columns (1) and (3) show that, in the low reputation group, the negative sentiment of stock message boards (*Neg-sent1/ Neg-sent2*) are significantly positive at the level of 1%, while Columns (2) and (4) show that in the high reputation group, the coefficients of independent variables are not significant. The above results show that firms with poor reputations are more strongly motivated to improve or repair their corporate image through ESG greenwashing and also mean that two types of media (traditional news media and social media) jointly influence ESG greenwashing.

Further Analyses

In addition to the myopia of managers and their motivation for impression management, a lack of external supervision is a deep-rooted reason for the frequent occurrence of ESG

Table 14 Cross-sectional tests of reputation

Variable	*Greenwashing*		*Greenwashing*	
	(1) Reputation-low	(2) Reputation-high	(3) Reputation-low	(4) Reputation-high
Neg-sent1	0.042***	0.021		
	(2.818)	(1.296)		
Neg-sent2			0.227***	0.122
			(2.640)	(1.387)
Growth	0.083**	0.025	0.084**	0.025
	(2.120)	(0.774)	(2.125)	(0.791)
Size	− 0.121**	− 0.064	− 0.122**	− 0.065
	(− 2.385)	(− 1.154)	(− 2.395)	(− 1.168)
Lev	0.368*	0.241	0.366*	0.241
	(1.749)	(1.116)	(1.743)	(1.119)
Exeshare	− 0.239	− 0.574*	− 0.231	− 0.572*
	(− 0.699)	(− 1.946)	(− 0.674)	(− 1.944)
ROA	0.376	− 0.021	0.373	− 0.017
	(0.987)	(− 0.058)	(0.978)	(− 0.048)
Dual	0.079	− 0.024	0.079	− 0.024
	(1.362)	(− 0.446)	(1.352)	(− 0.435)
Indep	− 0.107	0.664	− 0.098	0.663
	(− 0.215)	(1.273)	(− 0.196)	(1.272)
Board	0.169	0.171	0.169	0.171
	(1.064)	(0.789)	(1.062)	(0.789)
Top1	0.225	0.261	0.228	0.263
	(0.689)	(0.790)	(0.700)	(0.796)
SOE	− 0.168	− 0.052	− 0.168	− 0.053
	(− 1.578)	(− 0.345)	(− 1.589)	(− 0.348)
Age	− 0.272***	− 0.289***	− 0.272***	− 0.290***
	(− 2.994)	(− 2.818)	(− 2.999)	(− 2.826)
Pollute	0.326*	0.174	0.326*	0.174
	(1.712)	(0.858)	(1.715)	(0.863)
Diff	0.045**		0.051***	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	6048	4838	6048	4838
Adj.R2	0.399	0.355	0.399	0.355

This table presents the cross-sectional tests of reputation. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

greenwashing. Institutional investors, auditors and local environmental regulation, as important external supervisory forces in the capital market, can effectively improve the market information disclosure environment. Therefore, in this section, we further examine whether the effect of social media pressure on ESG greenwashing varies with institutional investors' shareholding, audit quality and local environmental regulation intensity.

Institutional Investors' Shareholding

Compared to small investors, institutional investors have a stronger professional capacity, governance motivation and influence, which enables them to effectively identify and constrain manager's myopic and self-interested behaviour. At the same time, institutional investors adhere to the concept of value investment and pay more attention to

Table 15 Cross-sectional tests of institutional investors' shareholding

Variable	*Greenwashing*		*Greenwashing*	
	(1) Shareholding-low	(2) Shareholding-high	(3) Shareholding-low	(4) Shareholding-high
Neg-sent1	0.049***	0.036*		
	(3.343)	(1.845)		
Neg-sent2			0.278***	0.189
			(2.912)	(1.555)
Growth	0.047	0.127***	0.048	0.127***
	(1.564)	(3.757)	(1.635)	(3.770)
Size	− 0.109***	− 0.139**	− 0.110***	− 0.141**
	(− 3.181)	(− 2.441)	(− 3.204)	(− 2.478)
Lev	0.326	0.218	0.323	0.220
	(1.710)	(0.805)	(1.704)	(0.810)
Exeshare	− 0.415*	− 0.431	− 0.409*	− 0.425
	(− 1.785)	(− 0.550)	(− 1.751)	(− 0.547)
ROA	0.070	0.246	0.068	0.244
	(0.218)	(0.711)	(0.212)	(0.711)
Dual	0.086	0.032	0.085	0.032
	(1.423)	(0.740)	(1.420)	(0.750)
Indep	0.301	0.277	0.303	0.275
	(0.647)	(0.699)	(0.652)	(0.692)
Board	0.042	0.431**	0.041	0.429**
	(0.238)	(2.126)	(0.232)	(2.113)
Top1	0.281	0.363*	0.287	0.366*
	(0.997)	(1.935)	(1.020)	(1.961)
SOE	− 0.013	− 0.277***	− 0.015	− 0.276***
	(− 0.157)	(− 3.525)	(− 0.186)	(− 3.503)
Age	− 0.366	0.306	− 0.370	0.304
	(− 0.496)	(1.520)	(− 0.503)	(1.507)
Pollute	0.195*	0.425**	0.195*	0.425**
	(2.004)	(2.543)	(2.001)	(2.543)
Diff	0.057*		0.046**	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	5208	5332	5208	5332
Adj.R2	0.675	0.667	0.675	0.667

This table presents the cross-sectional tests of institutional investors' shareholding. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

information pertaining to long-term value than to short-term performance (Stein, 1989). Moreover, the capital and information advantages of institutional investors can play a role in collaborative governance under the current popularity of ESG to enhance ESG performance and the transparency of ESG information (Yu & van Luu, 2021). However, the strength of institutional investors' supervision largely depends on their shareholding ratio in the company and the proportion of the investment in their portfolio (Fich et al., 2015). Therefore, an increase in institutional investors' shareholding ratio should help restrain ESG greenwashing induced by social media pressure.

To test our conjecture, we partition our sample based on the median value of the shareholding ratio of institutional investors (*Inshare*) in an industry year. The regression results are shown in Table 15. It can be seen that the coefficients of

Table 16 Cross-sectional tests of audit quality

Variable	*Greenwashing*		*Greenwashing*	
	(1) Non-Big 4	(2) Big 4	(3) Non-Big 4	(4) Big 4
Neg-sent1	0.041***	0.016		
	(3.551)	(0.563)		
Neg-sent2			0.236***	0.050
			(3.668)	(0.286)
Growth	0.056**	0.020	0.057**	0.018
	(2.221)	(0.174)	(2.260)	(0.161)
Size	− 0.098**	0.347**	− 0.099**	0.347**
	(− 2.432)	(2.226)	(− 2.470)	(2.220)
Lev	0.354**	− 0.662	0.355**	− 0.663
	(2.237)	(− 1.168)	(2.244)	(− 1.170)
Exeshare	− 0.560**	− 0.486	− 0.554**	− 0.478
	(− 2.276)	(− 0.536)	(− 2.251)	(− 0.529)
ROA	0.022	1.106	0.026	1.103
	(0.078)	(0.956)	(0.092)	(0.954)
Dual	0.042	0.168	0.042	0.168
	(0.958)	(1.271)	(0.968)	(1.275)
Indep	0.619	− 1.145*	0.621	− 1.150*
	(1.471)	(− 1.689)	(1.478)	(− 1.691)
Board	0.225	0.190	0.224	0.189
	(1.583)	(0.585)	(1.577)	(0.581)
Top1	0.086	− 0.764	0.094	− 0.769
	(0.371)	(− 1.135)	(0.406)	(− 1.142)
SOE	− 0.133	− 0.274	− 0.134	− 0.275
	(− 1.465)	(− 0.862)	(− 1.481)	(− 0.869)
Age	− 0.310***	− 0.181	− 0.311***	− 0.183
	(− 4.318)	(− 1.144)	(− 4.331)	(− 1.151)
Pollute	0.255*	− 0.010	0.255*	− 0.010
	(1.906)	(− 0.022)	(1.908)	(− 0.021)
Diff	0.004^{***}		0.002^{***}	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	9566	1320	9566	1320
Adj.R2	0.377	0.467	0.377	0.467

This table presents the cross-sectional tests of audit quality. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

variable (*Neg-sent1/Neg-sent2*) are significantly positive at the level of 1% in the group with low institutional investor shareholding ratio, while this effect disappears in the group with high institutional investor shareholding ratio. This is consistent with our prediction that the effect of social media pressure on ESG greenwashing is mitigated for firms with higher shareholding ratios of institutional investors.

Audit Quality

As providers of third-party market information certification, external auditors can effectively improve the corporate information disclosure environment. High-quality auditors hold high professional ethics and exercise a higher level of independence. They not only play an important role in financial information authentication but can also be used as an important supplement for improving the efficiency of corporate governance. Therefore, we predict that high audit quality can suppress the impact of social media pressure on ESG greenwashing.

We capture audit quality using a dummy variable that indicates whether the firm is audited by a Big 4 audit firm (*Big 4*), and partition our sample into two subsamples: companies hiring Big 4 and those hiring non-Big 4. The regression results in Table 16 show that, in the non-Big 4 audit groups, there is a significant positive correlation between the negative sentiment of stock message boards and ESG greenwashing, while in the Big 4 audit groups, this significant relationship does not exist. This result is consistent with the theoretical expectation, and reflects the important role of high-quality audit in the supervision of ESG greenwashing.

Local Environmental Regulation

The government plays a monitoring role to scrutinise to what extent firms act in an environmentally responsible manner (Ren et al., 2022). Effective environmental regulation can strengthen the stakeholder relationship between the government and firms and stimulate corporate green behaviour (Li et al., 2023; Ren et al., 2023). In addition, under pressure from local environmental regulation, firms must bear fines and reputation damage caused by inaction or greenwashing, and are thus more likely to actively carry out environmental responsibility practices and improve ESG performance. Therefore, the effect of social media pressure on ESG greenwashing is expected to be less pronounced for highly regulated firms.

To test this possibility, we use the proportion of investment in local environmental pollution control to GDP at the province level (*ER*) to measure the local environmental regulation strength level and explore the heterogeneous effects. Columns (1) to (4) in Table 17 present the results regarding local environmental regulation. The results show that social media pressure plays a significant role in promoting ESG greenwashing for firms subject to only low levels of regulation (Columns (1) and (3)), but this is not significant for highly regulated firms (Columns (2) and (4)). The results are consistent with our prediction that the effect of social media pressure on ESG greenwashing is mitigated for highly regulated firms.

Table 17 Cross-sectional tests of local environmental regulation

Variable	*Greenwashing*		*Greenwashing*	
	(1) Regulation-low	(2) Regulation-high	(3) Regulation-low	(4) Regulation-high
Neg-sent1	0.044***	0.020		
	(3.248)	(1.189)		
Neg-sent2			0.235***	0.129
			(3.082)	(1.345)
Growth	0.033	0.090**	0.034	0.090**
	(0.977)	(2.365)	(0.987)	(2.378)
Size	− 0.051	− 0.093	− 0.052	− 0.094
	(− 1.006)	(− 1.622)	(− 1.033)	(− 1.638)
Lev	0.280	0.337	0.279	0.337
	(1.449)	(1.361)	(1.448)	(1.363)
Exeshare	− 0.642***	− 0.395	− 0.641***	− 0.388
	(− 2.617)	(− 0.693)	(− 2.620)	(− 0.681)
ROA	0.072	0.250	0.069	0.259
	(0.213)	(0.540)	(0.204)	(0.558)
Dual	0.033	0.065	0.033	0.065
	(0.647)	(0.905)	(0.649)	(0.906)
Indep	0.232	0.271	0.228	0.274
	(0.445)	(0.471)	(0.437)	(0.477)
Board	0.045	0.273	0.042	0.274
	(0.242)	(1.397)	(0.228)	(1.404)
Top1	0.348	0.018	0.350	0.024
	(1.227)	(0.051)	(1.235)	(0.067)
SOE	− 0.070	− 0.124	− 0.070	− 0.125
	(− 0.673)	(− 0.692)	(− 0.675)	(− 0.702)
Age	− 0.300***	− 0.194*	− 0.302***	− 0.194*
	(− 3.447)	(− 1.754)	(− 3.469)	(− 1.753)
Pollute	0.100	0.130	0.102	0.131
	(0.545)	(0.860)	(0.554)	(0.861)
Diff	0.024**		0.098*	
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	6545	4341	6545	4341
Adj.R2	0.378	0.391	0.377	0.392

This table presents the cross-sectional tests of local environmental regulation. Reported in parentheses are t values based on robust standard errors clustered at the firm level. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. All variables are defined in the Appendix

Conclusion and Discussion

Based on impression management theory, this paper verifies the mechanism of social media pressure on corporate hypocrisy. Evidence from Chinese A-share listed firms shows that the negative sentiment on stock message boards can induce firms to conduct ESG greenwashing by exacerbating managerial myopia. In addition, we find that listed firms facing greater market value pressure and poorer reputations have a stronger impression management motivation and a higher degree of ESG greenwashing. Institutional investors' shareholding, high audit quality and local environmental regulation can play an effective role in external supervision, and alleviate the effect of social media pressure on ESG greenwashing. From a practical perspective, our conclusion is also of great importance in terms of improving the construction of an ESG system in emerging markets. That is, institutional investors, auditors and local environmental regulators should jointly participate in the construction of the ESG system and improve the ESG information environment. Investors should pay attention to the authenticity of ESG information and be wary of ESG greenwashing.

Our conclusion confirms the view of previous theoretical literature that the inconsistency between CSR statements and behaviours—that is, corporate hypocrisy—seeks to cope with the conflicting needs of stakeholders (e.g. short-term performance and CSR image) (Aguinis & Glavas, 2012; Brunsson, 1993; Cho et al., 2015). However, it is contrary to the theoretical framework proposed by Lyon and Montgomery (2013). Lyon and Montgomery (2013) argue that social media can promote information communication, increase the probability of consumers and activists discovering corporate greenwashing, and ultimately reduce the incidence of corporate greenwashing. This is basically consistent with the analysis based on principal–agent theory in this paper. We believe that the key to the contrary empirical results lies in whether social media users effectively identify corporate greenwashing behaviours and have a strong willingness to punish greenwashing firms. If the answer is yes, social media should be able to inhibit corporate greenwashing. If the answer is not so positive—that is, if social media users have difficulty effectively identifying corporate greenwashing, or if they can do this but have no will to respond—the converse will hold, and this may strengthen the motivation of companies to use ESG greenwashing for impression management in specific environments (such as when under negative public opinion), when ESG greenwashing may be an ideal low-risk behavioural strategy that can enhance the positive image of the firm. As Lyon and Montgomery (2013) point out, the key feature of their theoretical framework is that 'citizens and activists (imperfectly) monitor corporate environmental behaviour and decry corporate hypocrisy when they learn that a company does not 'walk the talk'. This can be seen as a necessary condition for its conclusion. In fact, unlike Lyon and Montgomery (2013), who analyse consumers and activists on Facebook and Twitter, our analysis focuses on small investors on stock message boards. Consumers and activists on Facebook and Twitter may be committed to exposing corporate greenwashing (mainly in the marketing field) and reaping the consequent reputation or other benefits of becoming opinion leaders. In other words, these people should be strong monitors of greenwashing and have a strong willingness to punish greenwashing. Small investors, however, are not. First, as discussed above, their information collection and processing capabilities are insufficient, which makes it difficult for them to identify the intrinsic motivation of corporate decisions (Malmendier & Shanthikumar, 2007) and accurately assess the environmental footprint of the firm (Lyon & Maxwell, 2011). In addition, ESG information is complex and more easily blurred than green product and service information in the marketing field. All these reasons may lead to small investors' failure to effectively identify ESG greenwashing. Second, small investors are short-sighted (Cremers et al., 2020). Even if they can identify corporate ESG greenwashing, it is not their optimal choice to expose it on social media. The reason is that, after exposure of corporate greenwashing, the stock price of the company will fall, and investors will face losses. In addition, the difference in institutional regulation is also an important reason for the difference between empirical and theoretical results. This paper carries out empirical analysis based on the Chinese context. Incomplete institutions and weak legal enforcement in China provide fertile ground for misbehaviour and window-dressing motivations (Du et al., 2023; Rong et al., 2024), and also greatly reduce any pressure arising from punishment for ESG greenwashing. As Lyon and Montgomery (2013) point out, 'If firms face little or no scrutiny, all types of firms have incentives to greenwash.' This is the situation that prevails in China.

This paper focuses on a number of key areas, the first of which is greenwashing in corporate ESG disclosure. Early literature pays attention to the false green promotion of firms to consumers or the general public in the marketing field (Jahdi & Acikdilli, 2009; Lyon & Montgomery, 2013). However, the greenwashing of firms is multidimensional and multi-level. Greenwashing not only refers to the disclosure of false environmental protection information to consumers and the public, but also may involve information window-dressing for investors, creditors and other stakeholders. Especially currently, ESG investment is highly respected. ESG information, as an important aspect of non-financial information, provides key references for investors and creditors to make decisions. It is thus a pity that prior literature ignores that catering to their needs may also act as an incentive for ESG greenwashing. The second area of focus is the influence of social media pressure. Although a stream of literature has analysed the internal and external factors that affect greenwashing (Kim & Lyon, 2015; Marquis et al., 2016; Testa et al., 2018; Zhang, 2022, 2023a, 2023c), empirical analysis based on public opinion pressure arising from social media is not common. Social media has emerged as an important channel for ordinary people to speak out, with unique advantages in shaping public opinion. In view of this, based on large sample data from China's largest stock message board (*EastMoney*), we examine the impact of negative sentiment on social media on ESG greenwashing. The conclusion contributes to a better understanding of the role of social media in inducing corporate hypocrisy. Our third area of focus is the behaviour of small investors. Protection of small investors is an eternal topic in the capital market. A large amount of literature explores the effectiveness of small investors in governance, but unfortunately, there are significant differences in conclusions. One literature stream notes that small investors can form certain constraints on managers or major shareholders to improve corporate governance (Ang et al., 2021; Chen et al., 2013; Fried et al., 2020; Li, 2021; Li & Zhang,

2023); another shows that small investors have limitations, which are not conducive to corporate governance and lead to chaos in corporate operations and governance (Belloc, 2013; Cremers et al., 2020; Malmendier & Shanthikumar, 2007). This paper further enriches the empirical evidence of the latter. However, it is worth noting that the above conflicts can arise from differences in group characteristics of small investors participating in governance and the specific issues of governance. Groups of small investors gathered through informal channels such as social media are more likely to expose their inherent information disadvantages and shortsightedness when facing various governance issues that are difficult to identify, such as ESG greenwashing, ultimately leading to negative outcomes. If the governance scenario is transferred to the formal shareholder meeting, or if small investors face clear governance issues such as expropriation or firm acquisition, a different story may emerge.

Finally, this paper has some limitations. First, for the construction of ESG greenwashing indicators, in the light of limitations in ESG information disclosure rating data, we use different rating agency data for ESG performance and ESG information disclosure. Although we have adopted standardised processing, because of different evaluation and scoring methods, the measurement of the degree of ESG greenwashing in this paper is still imperfect. Second, our empirical tests are based on the singular case of the Chinese market; given the influence of the policy, economic, environmental and cultural context, the research conclusions for different markets may differ. We hope to see more empirical evidence under different market environments in the future to enrich and improve the corresponding empirical conclusion. Third, because of space limitations, this paper does not discuss the internal governance factors affecting ESG greenwashing and the economic consequences of ESG greenwashing. More rich and in-depth research considering these could be carried out.

Appendix

Variable	Definition
Dependent variables	
Greenwashing	The deviation between normalised Bloomberg ESG disclosure score and normalised Huazheng ESG performance score
Greenwashing-wind	The deviation between normalised Bloomberg ESG disclosure score and normalised Wind ESG performance scores
Esgra	The Huazheng ESG performance score
Esgds	The Bloomberg ESG disclosure score
Independent variables	
Neg-sent1	$\frac{\text{Number of negative posts per year}-\text{Number of positive posts per year}}{\text{Number of negative posts per year}+\text{Number of positive posts per year}} \times$ Natural logarithm of total posts per year
Neg-sent2	(Number of negative posts per year + 1)/(Number of positive posts per year + 1)
Neg-sent3	(Number of negative posts per year + 1)/(Number of total posts per year + 1)
Neg-sent4	Natural logarithm of the annual number of negative posts
Other variables	
Growth	(Operating income of the period—operating income of the previous period)/ operating income of the previous period
Size	Natural logarithm of total assets
Lev	Total debt divided by total assets
Exeshare	Percentage of shares owned by executives
Roa	Net income divided by total assets
Dual	Dummy variable that equals one if the firm's CEO is also the chair of the board, and zero otherwise
Indep	Number of independent directors divided by the total number of directors
Board	Natural logarithm of the number of directors on the board
Top1	Percentage of shares owned by the largest shareholder
SOE	Dummy variable that equals one if a firm is a state-owned enterprise, and zero otherwise
Age	Natural logarithm of the number of years the firm has been listed
Pollute	Dummy variable that equals one if the firm is belongs to the heavily polluting industries, and zero otherwise
Baidu	The firm's Baidu search index
Loss	Dummy variable of loss, equaling one if the net income is negative and zero otherwise
Analyst	Natural logarithm of (1 + number of analysts following the firm)
GDP	Regional economic development level, measured as the natural logarithm of GDP at the province level
IE	Regional Internet development level, measured as the Internet penetration rate at the province level
Interaction	Dummy variable that equals one if there is an investor relationship management or investor consultation section on the official website homepage of firm, or an official interactive channel on WeChat and Weibo, and zero otherwise
BroadBand	Natural logarithm of the number of Internet broadband access users at the province level
Greenwashing_Dum	Dummy variable that equals one if *Greenwashing* is not a missing value, and zero otherwise
Readratio	Reading volume of posts per year /number of total posts per year
Exeage	The average age of a firm's executives

Variable	Definition
CutRD	Dummy variable that equals one if year *t*'s R&D expense is lower than that of year *t-1*, and zero otherwise
Index	Total myopic behaviour words*100/total MD&A words
TobinQ	Market value divided by book assets
Reputation	The number of negative media coverage of the firm in a year
Inshare	Percentage of shares owned by institutional investors
Big4	Dummy variable that equals one if the firm is audited by a Big 4 audit firm, and zero otherwise
ER	Regional environmental regulation intensity, measured as the proportion of investment in environmental pollution to GDP at the province level

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Declarations

Conflict of interest This paper does not involve any potential conflict of interest.

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